

DEEPHYPERION-UAV at the SBFT 2024 Tool Competition - CPS-UAV Test Case Generation Track

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ABSTRACT

Unmanned Aerial Vehicles (UAV) have showcased the potential for autonomous flights in real-world settings, sparking considerable interest across various application domains. Ensuring the robustness and safety of UAV operations in real-world scenarios necessitates a more comprehensive approach to testing, an aspect that has yet to receive adequate attention. In this paper, we proposed an automated test input generator for testing Unmanned Aerial Vehicles, named DEEPHYPERION-UAV that leverages the key advantages of Illumination search. DEEPHYPERION-UAV searches for diverse misbehaviour-inducing inputs spread across the cells of a map representing the feature space of the system.

KEYWORDS

Tool Competition, Software Testing, Test Case Generation

1 INTRODUCTION

We introduce DEEPHYPERION-UAV, a search-based approach for generating tests for Unmanned Aerial Vehicles (UAVs), submitted for participation in the CPS-UAV Test Case Generation Track at SBFT 2024 [2, 3]. The DEEPHYPERION-UAV technique applies the DeepHyperion-CS tool, previously introduced by Zohdinasab et al. [6] to the autonomous vision-based UAV navigation domain [1]. Its goal is to generate test cases that lead the UAV under test to fail by extensively exploring the feature space of the test input. It enables the identification of the highest-performing test cases, specifically those that provoke a misbehaviour or are in close proximity to misbehaving instances, distributed across the cells of a feature space representing the system. DEEPHYPERION-UAV outputs diversified critical scenarios for testing UAV systems through a feature map. The feature map serves as a representation of the feature space, defined by a set of N relevant dimensions of variation (i.e., the map axes, each corresponding to an input feature). On the feature map, test inputs find their placement according to their respective feature values. These maps serve as insightful tools, unveiling associations between feature value combinations and tests that resulted in misbehaviours. Hence, they indicate the likelihood of encountering a misbehaviour for each specific feature combination. Figure 1 shows an instance of a bi-dimensional feature map for a UAV system. This feature map is defined by the distance and angle

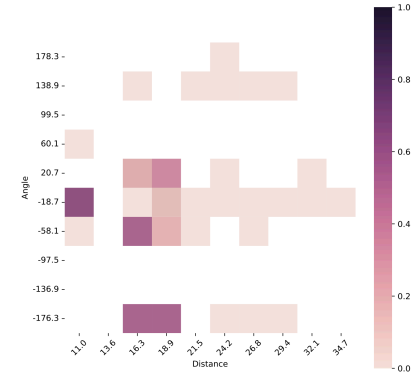


Figure 1: Feature map for a UAV navigation system. The axes quantify the distance and angle between two obstacles. The cells report the probability of exposing a misbehaviour for the corresponding feature value combinations, i.e., darker colors correspond to higher misbehaviour probabilities.

between two obstacles in the test scenario. Each combination of feature values is associated with a specific map cell, and the shading of each cell reflects the likelihood of encountering a misbehavior, where darker colors signify higher probabilities. In the example of Figure 1, the UAV system under test is likely to fail for obstacles positioned very close to each other when the drone fails to maintain a minimum safe distance of 1.5 m to the surrounding obstacles.

2 THE DEFINITION OF FEATURES

A crucial element of our approach is the choice of the dimensions that define the feature space of interest. In particular, the features should represent meaningful properties of the test scenarios, i.e. discriminative and interpretable properties of the inputs. In this study, we defined feature maps by considering high-level features that effectively characterize the UAV testing problem from the input viewpoint. Specifically, we adopted the following two features that were identified by applying the feature definition methodology proposed in the state-of-the-art approach [5]:

- *Obstacle Distance (Distance)*: measures the Euclidean distance between the center of two obstacles;
- *Obstacle Angle (Angle)*: measures the angle between the center of two obstacles;

3 THE DEEPHYPERION-UAV TECHNIQUE

The DEEPHYPERION-UAV uses an Illumination Search algorithm called Multi-dimensional Archive of Phenotypic Elites (MAP-Elites)



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Algorithm 1: DEEPHYPERION-UAV's Illumination Search

Input : B : execution budget
 $featurelist$: list of features
 $popsiz$: population size
 $rankselectionprob$: rank selection probability
 $rankbias$: rank bias

Output : M : feature map

```

1   $map\ M \leftarrow INITIALIZEMAP(featurelist);$ 
2   $population\ P \leftarrow GENERATEINITIALPOPULATION(popsiz);$ 
3  foreach  $ind \in P$  do
4     $EVALUATE(ind);$ 
5     $M \leftarrow UPDATEMAP(ind);$ 
6  end
7  while  $elapsedBudget < B$  do
8     $ind \leftarrow CS-RANKSELECTION(M, rankselectionprob,$ 
         $rankbias);$ 
9     $ind_{\mu} \leftarrow MUTATE(ind);$ 
10    $EVALUATE(ind_{\mu});$ 
11    $M \leftarrow UPDATEMAP(ind_{\mu});$ 
12    $ind \leftarrow UPDATECS(M);$ 
13 end
14 return ( $M$ )

```

proposed by J.B. Mouret and J. Clune [4] and improved it by introducing a rank selection mechanism based on the Contribution Score (CS) of the individuals [6].

1) *Search Algorithm:* Algorithm 1 illustrates the high-level steps of the DEEPHYPERION-UAV. At the start, a feature map M is defined with N dimensions (where N is the number of features in the feature list). The initial population of $popsiz$ is generated randomly (line 2). The individuals are evaluated by an Evaluate function, which computes the fitness function and the features' values of the individual and places it into M with the UPDATEMAP function (lines 3-6). After the initialization, the algorithm performs the main evolutionary loop (lines 7-13) until it reaches the budget B . At each iteration, an individual ind is selected from the feature map using the rank selection mechanism (line 8) to be mutated. The new individual ind_{μ} is generated by the mutation process and evaluated by the EVALUATE function. Each individual occupies a specific cell on the map based on its features' values. However, the map is updated with the new mutated individual ind_{μ} only if either the fitness value of ind_{μ} is higher than the individual currently occupying that cell or the cell is empty (line 11). Then, the algorithm updates the CS of the parent individual ind according to its contribution to the map's exploration (line 12). Finally, when the termination condition is satisfied the algorithm outputs the final map M together with all generated diverse high-performing test cases.

2) *Input Representation:* For the vision-based UAV navigation problem, we model the input as the location, size, and rotation of the two obstacles.

3) *Initial Population:* We randomly place two obstacles at positions in the mission area in such a way that there is no overlap.

4) *Fitness Function:* We define the fitness function such that it quantifies how close the drone is to exhibiting a misbehaviour. We

calculate the fitness of a test as min_dist that indicates the minimum distance of the trajectory to any of the obstacles. DEEPHYPERION-UAV aims to minimise this fitness function causing the drone to get closer to obstacles or lead to a collision.

5) *Selection:* Individuals are selected based on a probability proportional to their rank, where higher-ranked individuals have greater chances of being selected. The ranking is determined by the CS-RankSelection, which relies on the Contribution Score (CS). This score reflects an individual's contribution to exploration, considering whether it generated mutants that occupied previously empty cells or replaced existing individuals with better ones. The $rankbias$ parameter affects the algorithm by biasing the selection towards individuals with higher or lower rank.

6) *Mutation:* To search for the optimal obstacle solution, we use the following mutation operators for any of the obstacles individually:

- **Move:** moves the obstacle from the previous location in the simulated world by Δx or Δy .
- **Resize:** resizes the obstacle by Δl , Δw , Δh .
- **Rotate:** rotates the obstacle around its geometric center by Δr degrees.

4 EXPERIMENTAL RESULTS

In the competition [3], DEEPHYPERION-UAV has been applied for the generation of challenging scenarios for autonomous vision-based UAV navigation system. DEEPHYPERION-UAV ranked 4th in the ranking of the effectiveness of the different test case generators in the competition. In particular, detailed results showed that the DEEPHYPERION-UAV is effective in generating failure-inducing inputs associated with highly diverse features.

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